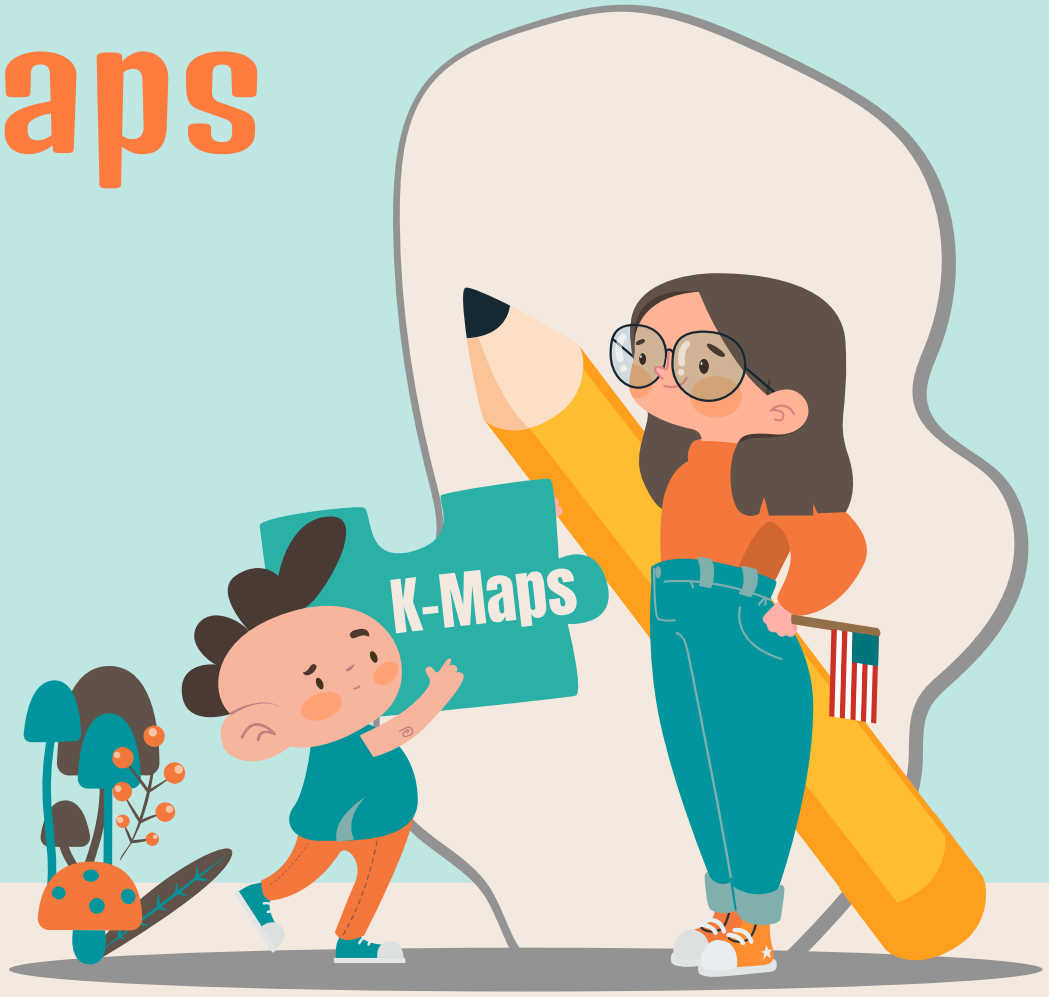
Karnaugh Maps

Part II

Lecture 7



04

Minimization using K-Map

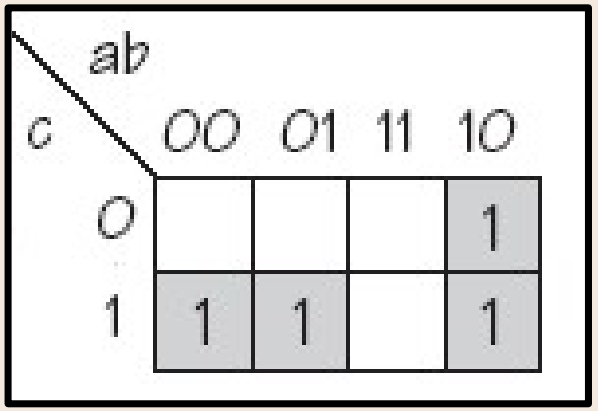


Minimization using Karnaugh map

SOP = $\bar{a}.\bar{b}.c + \bar{a}.b.c + a.\bar{b}.\bar{c} + a.\bar{b}.c$

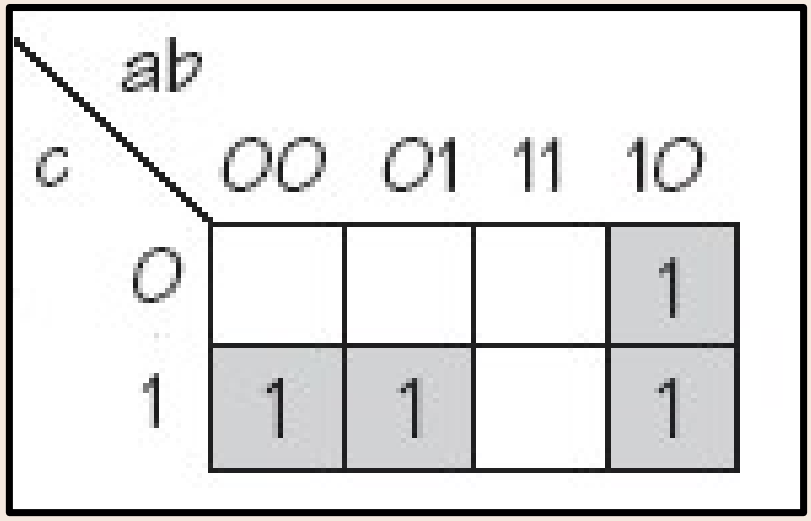
a	b	c	y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0





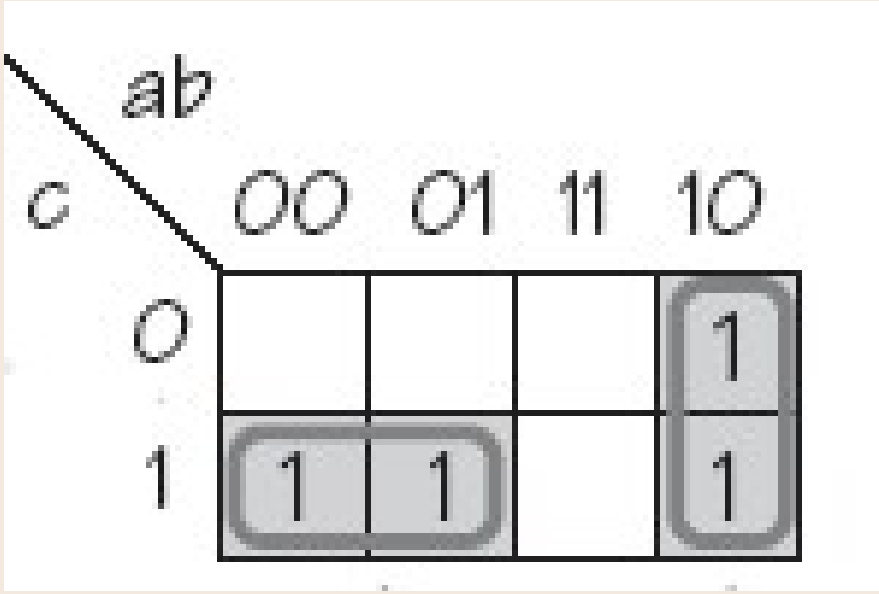
Minimization using Karnaugh map

- The minimization process is based on grouping the 1's inside the K-map.
- You can group adjacent 1's in the power of 2 only.
- Not in diagonal direction.

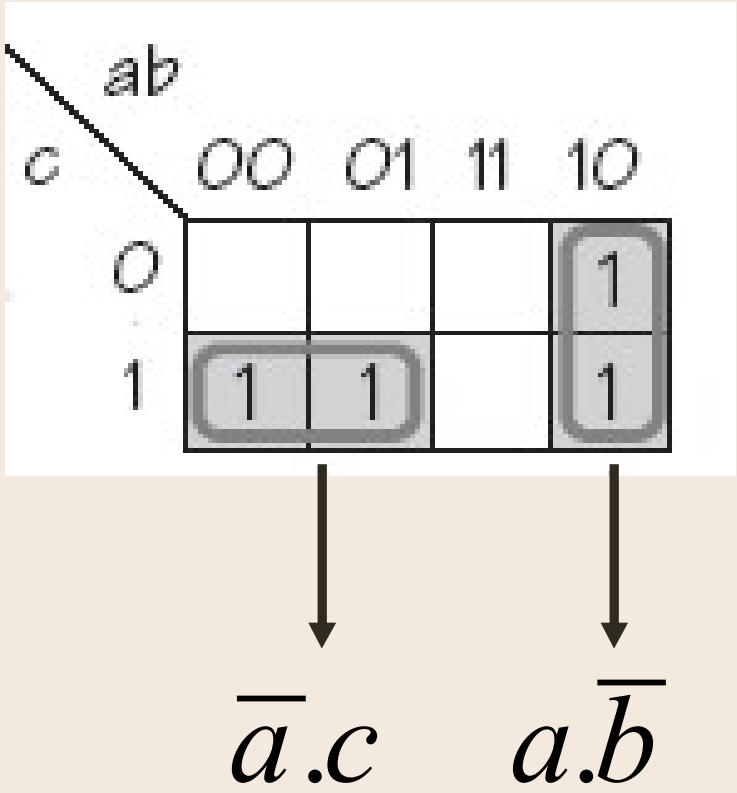


Minimization using Karnaugh map

- Group the 1's inside the K-map as follows.

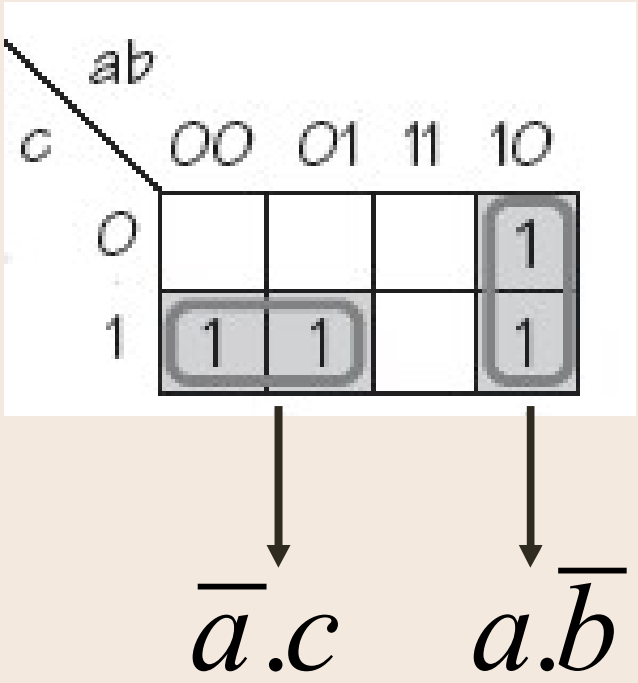


Minimization using Karnaugh map



Minimization using Karnaugh map

- Group the 1's inside the K-map as follows.



Minimized expression

$$y = \overline{a}.c + a.\overline{b}$$

Compare this expression with the original SOP expression

$$\overline{a}.\overline{b}.c + \overline{a}.b.c + a.\overline{b}.\overline{c} + a.\overline{b}.c$$

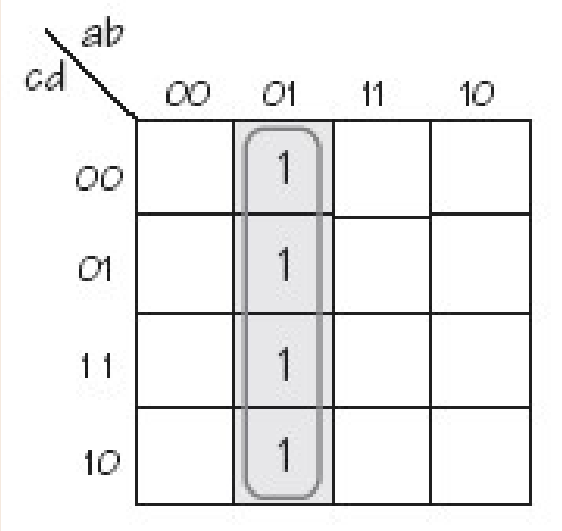
05

Grouping inside K-Maps

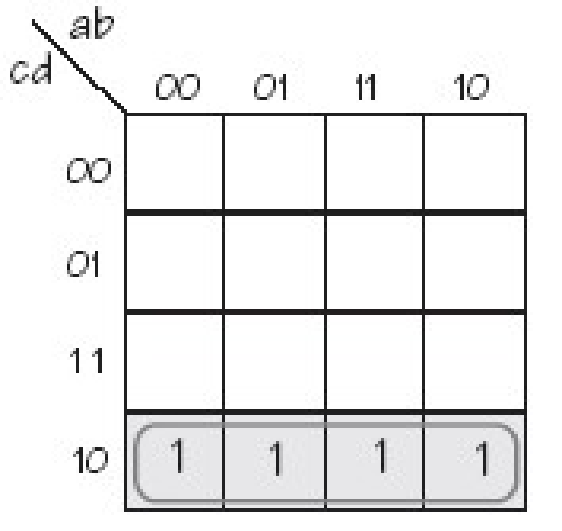


Grouping inside the K-map

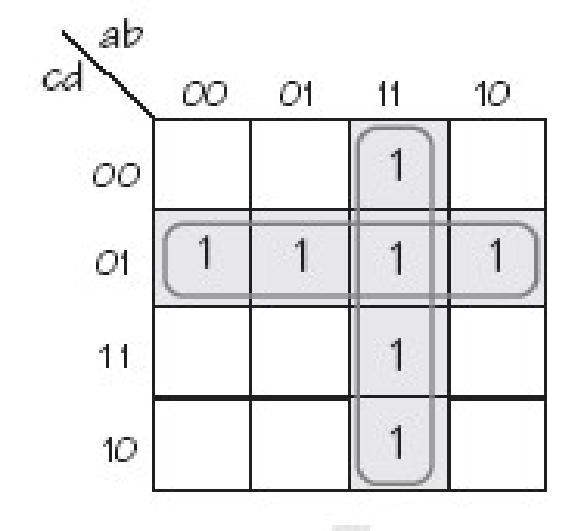
Case 1



Case 2

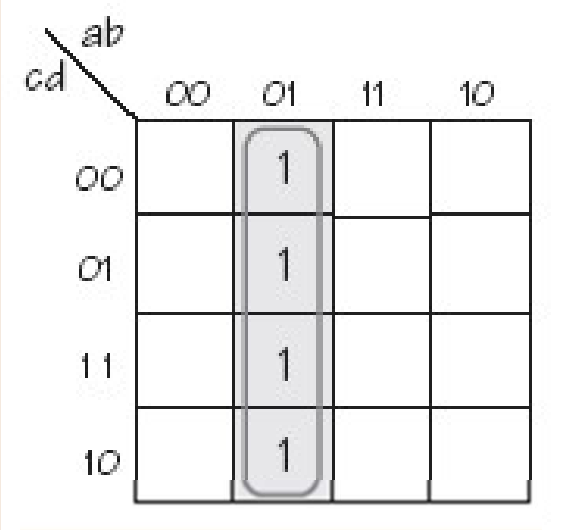


Case 3



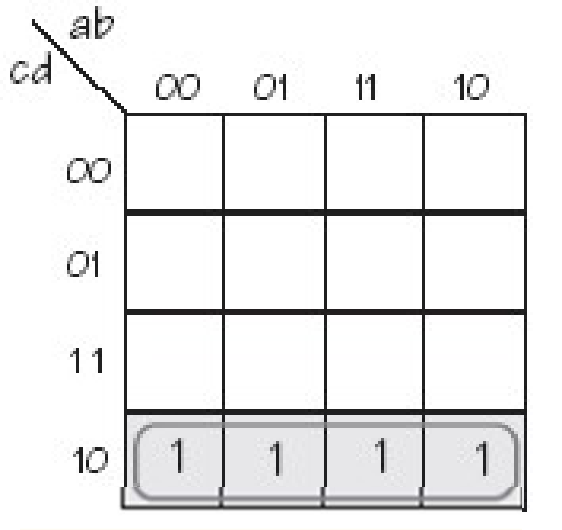
Grouping inside the K-map

Case 1



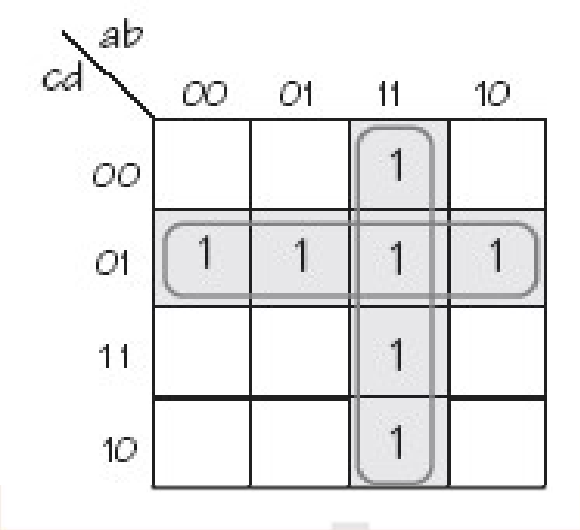
$$y = \bar{a}.b$$

Case 2



$$y = c.\bar{d}$$

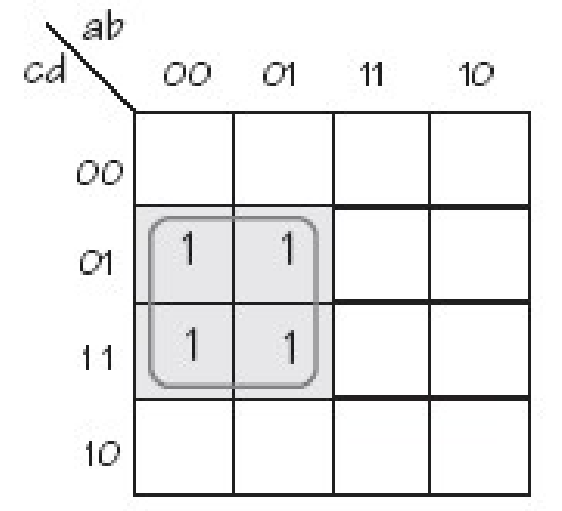
Case 3



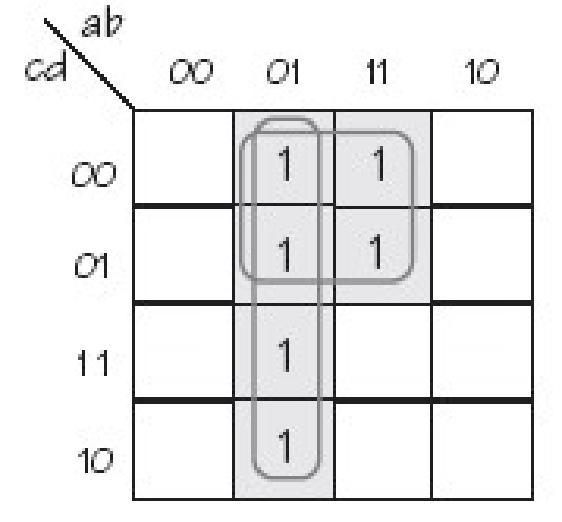
$$y = a.b + \bar{c}.d$$

Grouping inside the K-map

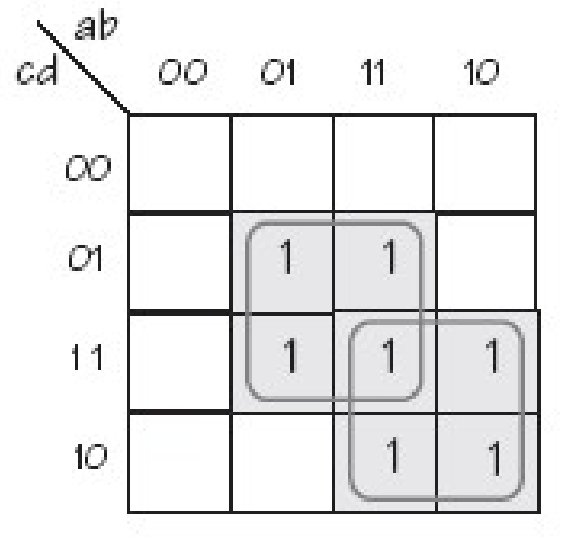
Case 4



Case 5

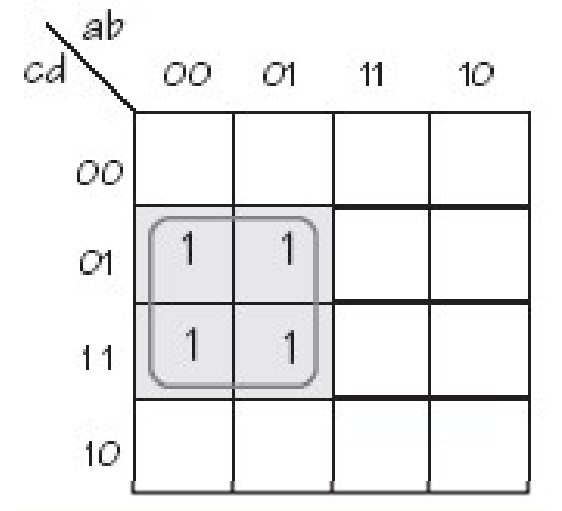


Case 6



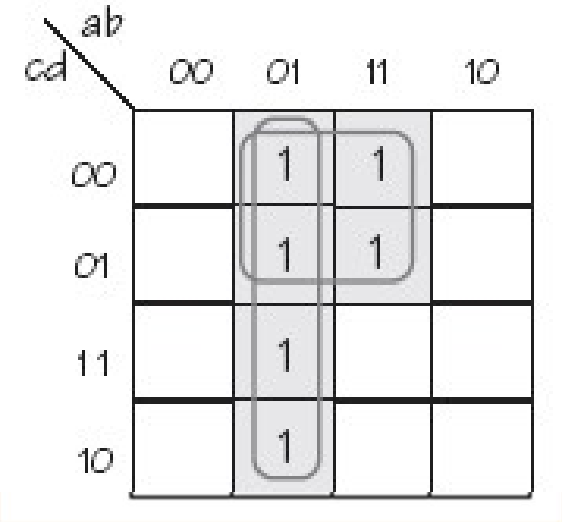
Grouping inside the K-map

Case 4



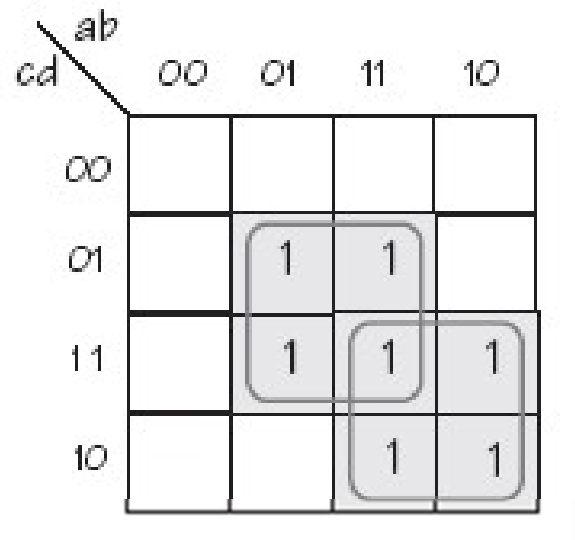
$$y = \bar{a}.d$$

Case 5



$$y = \bar{a}.b + b.\bar{c}$$

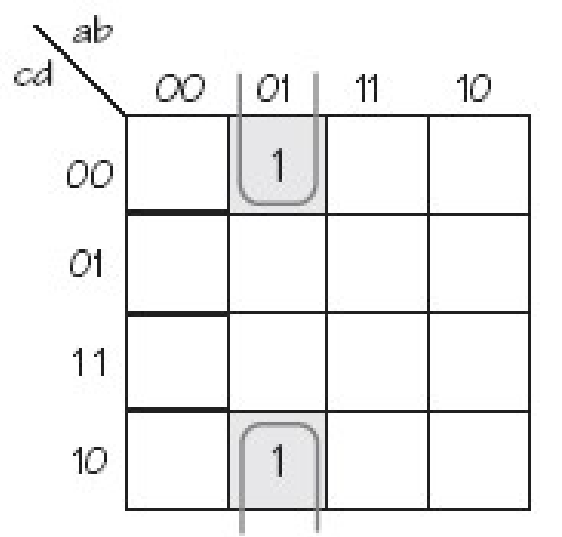
Case 6



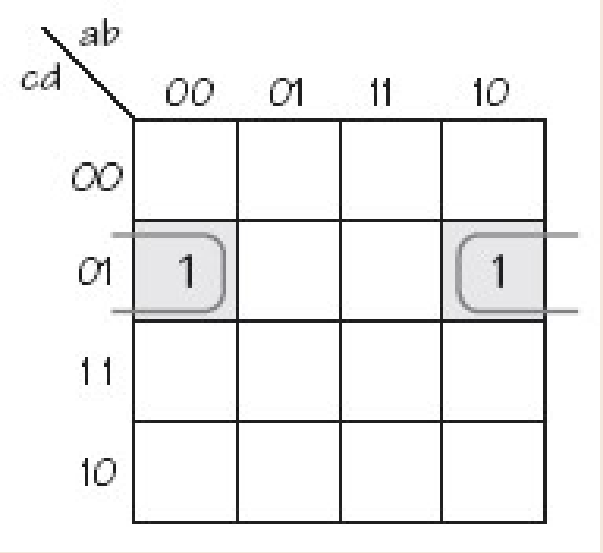
$$y = b.d + a.c$$

Grouping inside the K-map: Map Rolling Method

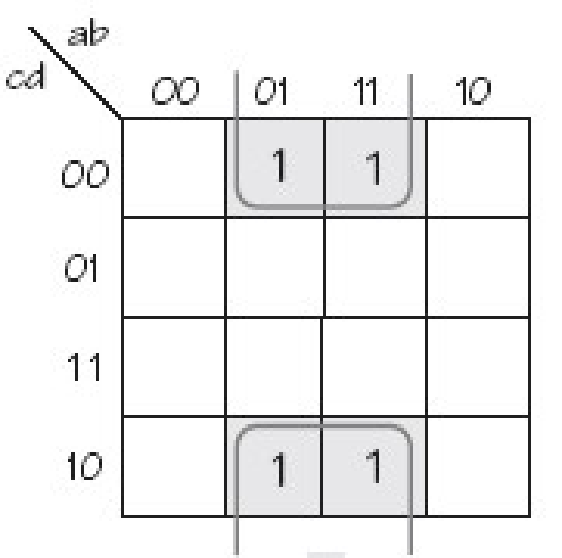
Case 7



Case 8

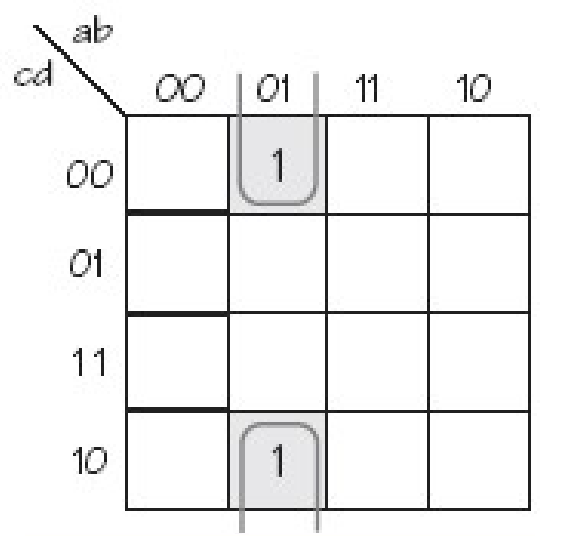


Case 9



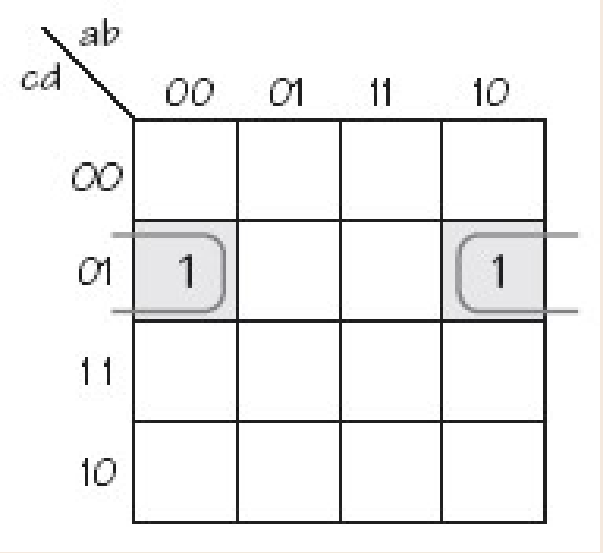
Grouping inside the K-map: Map Rolling Method

Case 7



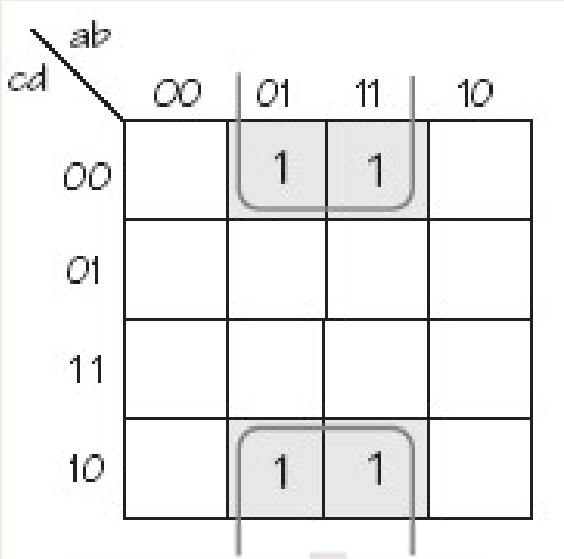
$$y = \bar{a}.b.\bar{d}$$

Case 8



$$y = \bar{b}.\bar{c}.d$$

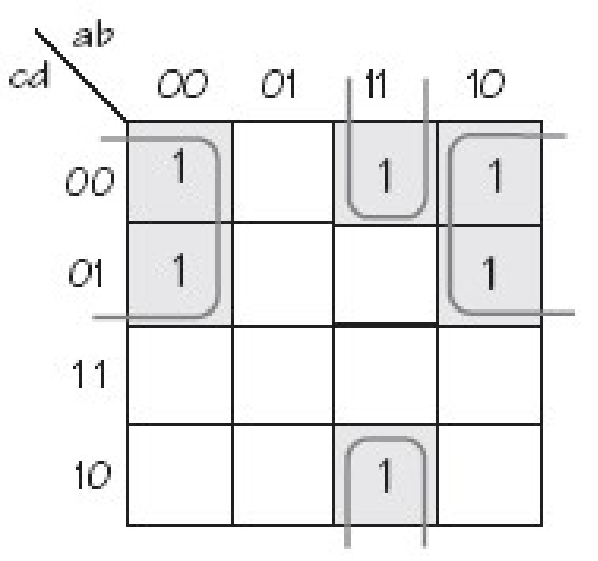
Case 9



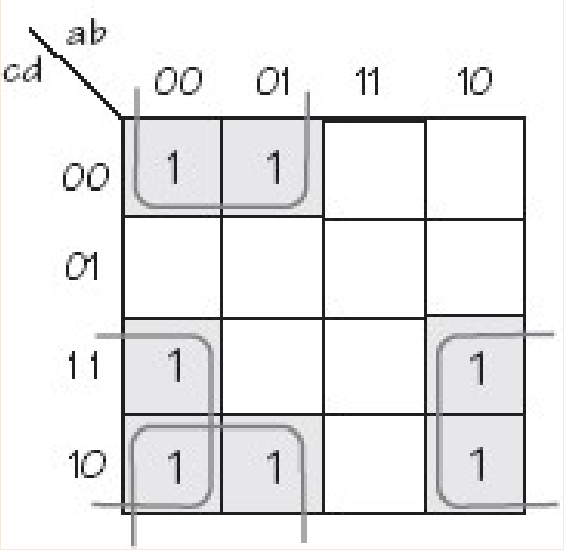
$$y = b.\bar{d}$$

Grouping inside the K-map

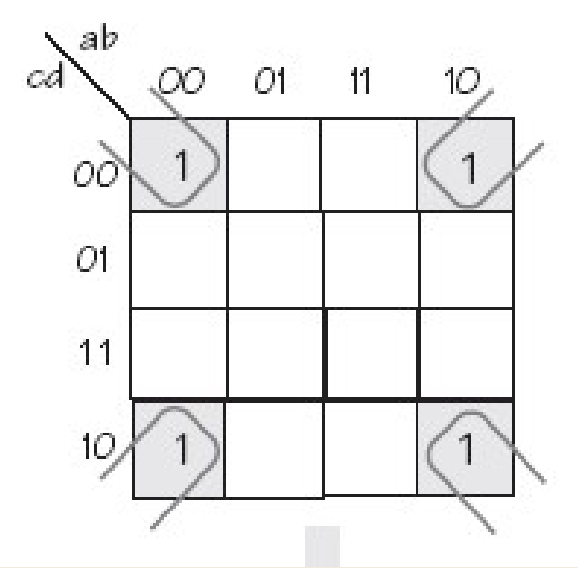
Case 10



Case 11

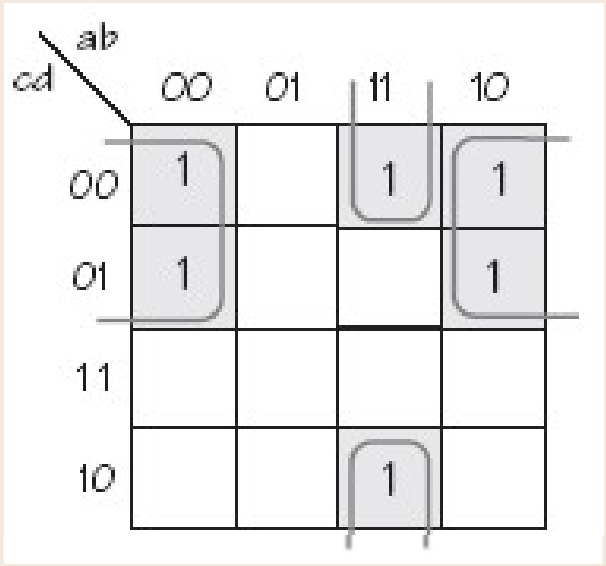


Case 12



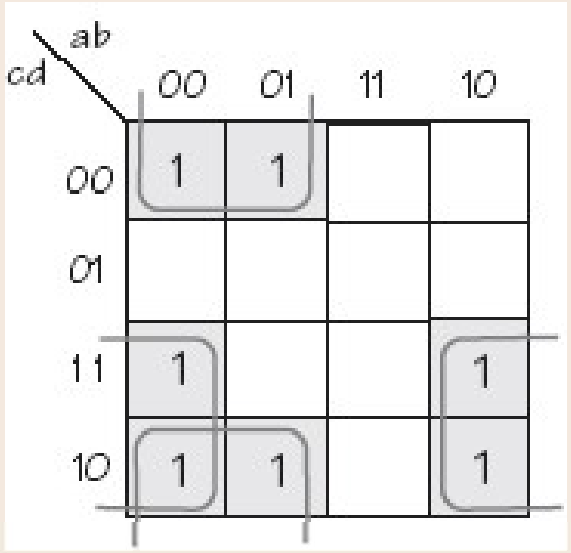
Grouping inside the K-map

Case 10



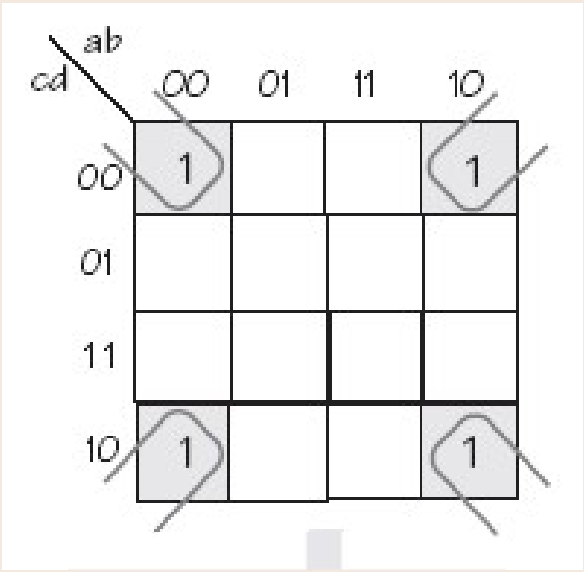
$$y = \bar{b}.\bar{c} + a.b.\bar{d}$$

Case 11



$$y = \bar{a}.\bar{d} + \bar{b}.c$$

Case 12



$$y = \bar{b}.\bar{d}$$

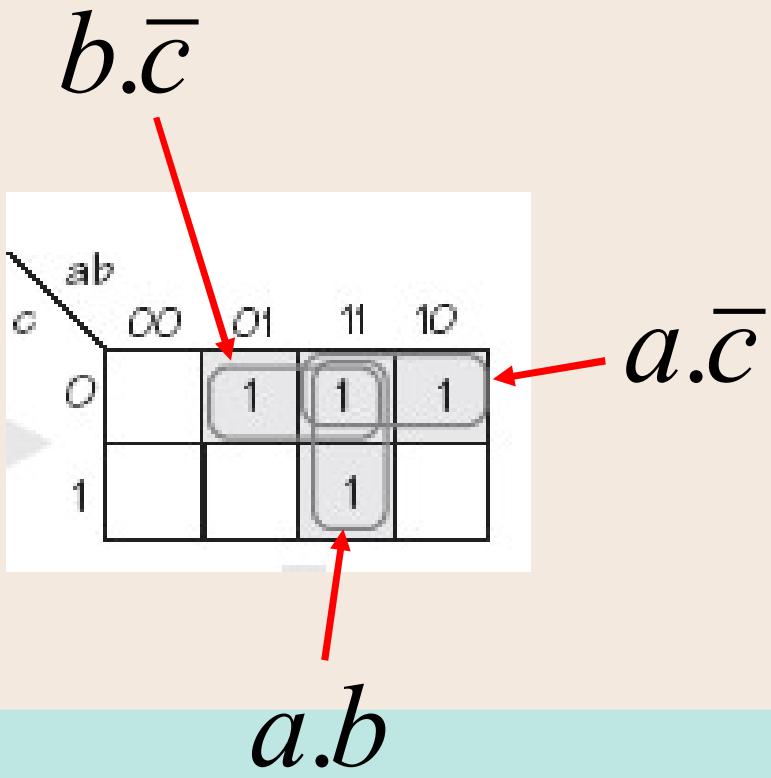
Question: Write down this function in SOP form.

a	b	c	y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



Find the Minimized Expression using K-Map

- Group the 1's inside the K-map as follows.



Minimized expression

$$y = a.b + b.\bar{c} + a.\bar{c}$$

Compare this expression with the original SOP expression

Question: Obtain the minimized logic expression of the following Boolean function using K-Maps.

inputs				out
A	*B*	*C*	*D*	*Y*
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1



THANKS!

Have a nice day!

